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Methods and approaches to support Indigenous water planning: An example from the Tiwi Islands, Northern Territory, Australia

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SUMMARY

Indigenous land owners of the Tiwi Islands, Northern Territory Australia have begun the first formal freshwater allocation planning process in Australia entirely within Indigenous lands and waterways. The process is managed by the Northern Territory government agency responsible for water planning, the Department of Natural Resources, Environment, The Arts and Sport, in partnership with the Tiwi Land Council, the principal representative body for Tiwi Islanders on matters of land and water management and governance. Participatory planning methods ('tools') were developed to facilitate community participation in Tiwi water planning. The tools, selected for their potential to generate involvement in the planning process needed both to incorporate Indigenous knowledge of water use and management and raise awareness in the Indigenous community of Western science and water resources management.

In consultation with the water planner and Tiwi Land Council officers, the researchers selected four main tools to develop, trial and evaluate. Results demonstrate that the tools provided mechanisms which acknowledge traditional management systems, improve community engagement, and build confidence in the water planning process. The researchers found that participatory planning approaches supported Tiwi natural resource management institutions both in determining appropriate institutional arrangements and clarifying roles and responsibilities in the Islands' Water Management Strategy.

1. Introduction

This article reports research and outcomes from the Water Planning Tools (WPTs) project, designed to facilitate Indigenous[2] community engagement in a water planning process on the Tiwi Islands in the Northern Territory (NT), Australia. The Tiwi water planning process is the first formal process for a freshwater system entirely within Aboriginal lands and waterways in Australia. It falls within the framework for water planning provided under the national blueprint for water reform, the National Water Initiative (NWI) (Council of Australian Governments, 2004) described more fully elsewhere (Tan et al., 2012, this issue). The process is a partnership between the Tiwi Land Council (TLC), the representative body for land and water management for the Tiwi Islands, and the NT government agency responsible for water planning, the Department of Natural Resources, Environment, The Arts and Sport (NRETAS). The broad objective of the research was to develop approaches and mechanisms—referred to here as 'planning tools'—to assist and support Tiwi landowners and their institutions in undertaking water planning in the context of contemporary water management in Australia. The aims were to enhance awareness of the Tiwi planning process, to promote community engagement, and to build and augment social learning. It was anticipated that the planning tools would enable Tiwi people to make informed decisions about the future management of their water resources.

The challenge was to find ways to articulate Tiwi knowledge, aspirations and interests in water resources in the water management plan. To address this challenge, researchers from the Commonwealth Scientific and Industrial Research Organisation (CSIRO) and Griffith University worked closely with the TLC, community members, the Water Resources Branch of NRETAS, and the NT Power and Water Commission (PowerWater). Research goals, defined in an agreement between the WPT research team and the TLC, aimed to support Tiwi people in recording their knowledge and perspectives on water resources and features as well as future water demands; and to determine the cultural, social, economic and environmental impacts of water use and management options (Tiwi Land Council, 2009). The action research methodology is fully described in Mackenzie et al. (2012, this issue). Researchers worked in collaboration with Tiwi institutions and

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community members to iteratively identify research questions, implement and evaluate research practice.

2. Planning context and issues

In recent years state and federal governments and the National Water Commission have supported several projects exploring how Indigenous knowledge of water resources can be best accounted for in government water planning processes (Hamstead et al., 2008; Jackson et al., 2009, 2010; Jackson, 2006, 2009; Jackson and Morrison, 2007; McFarlane, 2004). However, water allocation planning on the Tiwi Islands is unique in Australia as the first formal water planning process that will apply to wholly Indigenous-owned lands. It was important therefore to use culturally appropriate arrangements for water resource decision-making. Such arrangements would require consultation, flexibility, creativity, and openness on the part of research partners and the Tiwi community.

A wide range of participatory processes is used internationally and in Australia in community development and Natural Resource Management (NRM). They include the video, photographic and audio recordings of story-telling, testimonies and activities (Walsh and Mitchell, 2002; Srinivasan, 1993; Wang and Burris, 1997); 'cultural mapping' methods of Indigenous sites and knowledge using geographic information systems (Harmsworth, 1998; Corbett et al., 2006; Bhattarya and Nitesh, 2004; Puri, 2007); scenario development and visioning (Wollenberg et al., 2000); field visits to Indigenous estates or 'country' (Haig et al., 2003; Lynch et al., 2010); and facilitated, community-based workshops (Whiting et al., 2007; Moran, 2004; Ens et al., 2012, p. 103). Australian Indigenous communities and their research partners are also using multimedia techniques, including satellite imagery and topographic maps, to produce interactive maps and spatial interfaces as participatory planning tools (Corbett et al., 2006; Charles Darwin University, 2008).

Many of these tools are designed to elicit and record Indigenous knowledge and aspirations for management. Our work addresses a gap in current research by investigating how participatory planning tools can be used not only to represent Tiwi Indigenous knowledge in culturally appropriate ways but also as a means of facilitating participation in the planning process, and its iterative design.

National water policy recognizes the need for Indigenous representation to address Indigenous water management objectives. It is clear that more needs to be done to consider the cultural and economic expectations of Indigenous communities (Jackson et al., 2012, this issue; National Water Commission, 2011). As Lane and McDonald (2005) note, 'the capacity of indigenous [sic] people to participate in planning processes is…a crucial factor in determining the extent to which planning outcomes reflect, at least in part, indigenous priorities.' By embedding planning tools within a formal water allocation process, this research can offer insights into participatory planning to support sustainable water resource management on Indigenous estates, and has implications for the realization of aspirations of Indigenous communities in Australia and elsewhere.

2.1. Biophysical – hydrological and ecological context

The Tiwi Islands of Melville and Bathurst are located 60 km north of Darwin, the capital of the Northern Territory. The two islands, formed between 18,000 and 7000 years ago when rising sea levels separated the islands from the mainland (Haig et al., 2003), are characterized by tropical monsoonal vegetation including savannah woodlands, monsoonal vine forests, wetlands and mangrove forest along the coast alternating with white sandy beaches and cliffs.

There are two regional aquifer systems on the islands. The geological formation identified as Van Dieman Sandstone is a regional, shallow unconfined aquifer that covers most of both islands (Haig et al., 2003, p.19), and supports the groundwater-dependent ecosystems that characterize Tiwi landscapes. The second system is a deep aquifer of sandstone confined within impervious claystone and siltstone.

Surface water features include soaks, springs, streams/rivers, billabongs, swamps and waterholes dependent on groundwater recharged by seasonal rainfall during the wet season between October and April (Haig et al., 2003). The rate of flow and morphology of these features is contingent on seasonal fluctuations in rainfall and transpiration. All freshwater springs are maintained through the dry season by groundwater (Haig et al., 2003). There are both perennial and ephemeral features of particular significance to Tiwi Islanders. For example, 176 plant and 14 animal species dependent upon freshwater for survival are valued for traditional use (Tiwi Land Council, n.d. (a)).

Under customary law Tiwi Islanders have management responsibilities for all lands and waters. More importantly, from a Tiwi perspective, there is a strong connection between land, water and the associated plants and animals, such that the persons' identity is connected to the ownership, allocation and expression of land and natural resources.

2.2. Social and economic context

In 2006 the estimated residential population of the Tiwi Islands was 2512 people, of whom 92.4% were Indigenous (URS, 2007) drawn mainly from eight traditional owner groups. The majority of people live in the Bathurst Island community centers or townships of Nguiu (now known as Wurrumiyanga) and Wurankuwu (also known as Rangku), and at Pirlangimpi and Milikapiti on Melville Island (see Fig. 1).

Representatives from each of the eight traditional owner groups make up the Land Council, the native title representative body with overarching natural resource management and governance responsibilities. Islanders participate on the Council based on their affiliations and responsibilities for owning and managing lands, waters and resources (Kate Hadden, pers. comm., 2009).

Currently there is not a high demand for the water resources of the Tiwi Islands. Consumptive water use includes water for irrigation of small-scale horticultural crops, public open spaces and sporting facilities, domestic use (drinking water and household use), road works, and mineral sands mining. The 2003 Tiwi Islands Water Study determined that water resources of high quality were adequate for existing demand (Haig et al., 2003), an analysis confirmed by an NRETAS technical assessment in May 2010 (Gray and Paiva, 2010; Schult, 2010).

However Tiwi Islanders are interested in exploring opportunities for economic development, therefore population expansion and future commercial horticulture and other developments will increase demand on water resources. As part of the project the researchers canvassed Tiwi Islanders' interest in initiatives such as opportunities for new enterprises, future development and growth stimulated by community leasing and the NT 'Growth Towns' strategy (Tan et al., 2010) which would potentially expand small business and tourism development and therefore increase water use.

Through past collaborations with NRETAS (Haig et al., 2003) and the recent partnership with the NT government the Islanders have demonstrated a commitment to planning and managing their water resources sustainably (John Hicks, pers. comm., 2009). Both past involvement and demonstrated interest mark out the Tiwi

Fig. 1. Map of towns and outstations of the Tiwi Islands.

Islanders as interested partners in a trial to refine engagement methods and to elicit matters of importance in water planning. Lessons learned from this research process will assist NT and other Australian water planners to address and acknowledge Indigenous interests and systems.

2.3. Planning issues

The NT government manages its water resources through a regulatory framework that includes the *Water Act* (NT) 1992 which provides for the investigation, allocation, use, control, protection, management and administration of water resources, the *Water Regulations* (NT), and a series of water allocation plans. Statutory water allocation plans can only be developed in areas that have been declared as a Water Control District under the *Water Act* (NT). The Tiwi Islands is currently not a declared Water Control District.

The NT government is committed to achieving the targets, including statutory water planning, set by the federal government's water reform blueprint, the National Water Initiative (NWI). The NWI processes aim to reconcile the diversity of interests in the allocation or sharing of water for different purposes within a region according to collaboratively agreed rules. Planning on the Tiwi Islands is further complicated by federal legislation which gives statutory authority under the *Aboriginal Land Rights* (*Northern Territory*) *Act* 1976 (ALR Act) to the Tiwi Land Council to administer and manage land. Additional responsibilities arise under the federal *Native Title Act* 1993 and the *Northern Territory Pastoral Land Act* 1992.

While the ALR Act may grant inalienable communal land title to the Tiwi traditional owners, Australian land rights legislation generally makes no mention of Indigenous ownership or rights to freshwater or inland waters. Section 9, for example, of the *Water Act* (NT) declares that the Crown owns all surface and groundwater in the Northern Territory. Further, any grant of freehold title does not include minerals which, under the ALR Act, are defined to include water (Jackson and Altman, 2009, p. 30). Nevertheless Tiwi Islanders believe that freshwater on their Country belongs to them in accordance with their customary law, just as their land ownership rights are recognized by statutory law. Rights to use inland waters for the purposes of hunting, gathering, fishing and undertaking cultural and spiritual activity in the exercise or enjoyment of native title rights (Jackson and Altman, 2009) are recognized by the *Native Title Act* 1993 and NT legislation (*Northern Territory Parks and Wildlife Conservation Act* 1986).

In 2008 the TLC agreed to participate in a water planning process supported by the NT government and funded by the Australian Government through the National Water Commission. The NT government appointed Michael Schmid, the first Indigenous Water Planner in the Water Resources Branch of NRETAS, to assist in the development of a Tiwi Islands water allocation plan. NRETAS committed to supporting Tiwi landowners and the TLC to develop a water planning process that would be responsive to Tiwi governance and resource capabilities (Ian Lancaster, pers. comm., 2009), as well as providing a flexible approach to policy and legal issues associated with Tiwi water management under the NWI and NT government frameworks. These issues included, for example, whether Tiwi decision-makers would declare the Tiwi Islands a Water Control District, and consultation with Tiwi authorities on whether the eventual Water Management Strategy would have statutory or merely administrative force.

3. Planning challenges

The Water Planning Tools researchers identified a number of challenges facing the Tiwi water planning process, including designing the process and engaging Tiwi Islands stakeholders. The main stakeholders included the TLC, its Natural Resource Management (NRM) Committee, Tiwi elders who may not necessarily hold current positions on the Council, the Tiwi Land Rangers employed by the TLC to undertake land and water management activities, members of the Tiwi Islands Shire Council and the community. Other stakeholders included industry bodies such as Tiwi forestry and sand mining enterprises, and NT government environmental protection, agriculture and fisheries, infrastructure and planning, essential services and health and community services agencies.

A major challenge was that there was not yet any obvious issue with either the quantity or quality of water on the Tiwi Islands. Generally community members did not express major concerns for their water resources although the TLC and the NT government both recognized the importance of strategic planning to ensure future sustainable use of Tiwi water resources.

3.1. Research protocols and coordination

The WPT project methodology followed a three-stage process. In the first stage, WPT researchers worked with Tiwi research partners and NT water agency staff to determine the appropriate working arrangements for the project, and produce a context analysis for Tiwi water planning (Ayre, 2009). Secondly, they facilitated and participated in deliberations with these partners to select potential approaches to engaging the range of stakeholders. A research agreement between the WPT research team and the TLC defined the following goals: support the recording of Tiwi knowledge and perspectives on water resources and features as well as future water demands, and determine the cultural, social, economic and environmental impacts of water use and management options (see Tiwi Land Council, 2009). Thirdly, WPT researchers worked with water agency staff and Tiwi research partners to trial the chosen water 'planning tools'. These tools were selected by the researchers in consultation with the Water Planner, the Tiwi Land Rangers and the Environmental Secretariat of the Tiwi Land Council as the most appropriate to both engage Tiwi people in the water planning process and elicit Tiwi values and objectives for water planning. Researchers undertook participant observation at four meetings of the TLC and its NRM Committee and made over 20 visits to the Tiwi Islands to undertake the collaborative work of developing planning tools for the Tiwi water planning process. The total numbers of Tiwi people estimated to have been involved in the WPT project is over 100 including approximately 10–12 Tiwi Land Rangers, eight Tiwi Land Delegates, approximately 20–35 TLC members (at four formal meetings), Tiwi community members who attended the community workshops (see Section 4.1), and other interested people who attended the Tiwi Open Days and other informal interactions between WPT researchers and Tiwi people and their institutions.

The TLC's environmental policy promotes broad community involvement to achieve the Tiwi vision for the management of natural and cultural resources (Tiwi Land Council, n.d.(b)). Under customary law Indigenous land ownership, governance and decision-making acknowledge '…multiple bases for entitlement to access resources and … complex classificatory relationships locating all members of [Tiwi] society in relationships to each other and to resources' (Davies et al., 2008, p. 58). Thus the TLC required that the project's partner organizations should work in culturally appropriate ways to facilitate Tiwi visions and objectives for water use and management, engaging with Tiwi people in ways that elicited and reflected their knowledge and governance of water.

The complexity of administrative arrangements and responsibilities for water management on the Tiwi Islands was a challenge to the process and its embedded WPT project. The TLC and its Natural Resource Management Committee (the NRM Committee) are the primary institutions responsible for decisions about issues that affect Tiwi lands, waters and communities. In the planning process, these institutions worked with NRETAS, CSIRO and Griffith University personnel to provide ongoing guidance and approval for consultation mechanisms and for the development of water use and management options consistent with established Tiwi governance and procedural protocols.

Another challenge was the inclusion of the multiple institutions, government agencies and service providers with responsibilities for Tiwi water resources. Water supply is the responsibility of PowerWater which provides electricity, water and sanitation services to Indigenous communities (PowerWater, 2008, p. 2). During the Tiwi water planning process, PowerWater began its own water planning program working with communities to improve their water use, particularly in those Indigenous communities identified as 'water stressed' (PowerWater, 2008). To minimize potential confusion over water related roles and responsibilities, the researchers worked with NRETAS and PowerWater to coordinate engagement and communication with Tiwi communities and the TLC.

The WPT researchers developed protocols for engagement with the TLC, beginning with an environmental studies and research access agreement (Tiwi Land Council, 2009). This agreement formally sets out the relationship between parties and elaborates their responsibilities. In 2009 The TLC and NRETAS signed a Memorandum of Understanding to guide their partnership and the implementation of Tiwi water planning.

Given the unique context of the Tiwi water planning process, NRETAS and the researchers agreed to communicate regularly with the TLC and to spend time on developing appropriate ways of working with the Tiwi community.

3.2. Tool requirements

Tiwi Islanders, as with many other Indigenous peoples, have extensive and detailed knowledge of water resources derived from their custom, law and practice (Matsuyama and Haig, 2003; Jackson, 2005; Hiscock, 2008). For the Tiwi water planning process all parties needed to work with concepts and knowledge/s based in both Western scientific and Tiwi traditions. Most parties had little experience of the law behind each others' water management policies. We therefore sought to trial engagement methods that elicited and credited Tiwi values and water governance while also facilitating learning about Western scientific concepts such as the water cycle and hydrological systems.

The planning tools also needed to engage the Tiwi communities, decision-makers and institutions over a sufficiently long timeframe to raise awareness and understanding of the issues. Adequate time would allow all parties to engage with the project at a realistic pace, providing each other with explanations of planning and research techniques, and periodic debriefings (Jackson and O'Leary, 2006, p. 68).

4. Tool trials

From a context analysis of Tiwi water planning (Ayre, 2009) and in discussion with NRETAS and the TLC, we chose to focus on the following requirements:

- eliciting information to identify the uses, values and objectives Tiwi groups have for water and water management;
- cross-communicating Western scientific and Tiwi knowledges about water and hydrological systems; assessing the impacts of developing and managing water resources so that Tiwi Islanders can make informed decisions about water management.

Participatory engagement methods and approaches, referred to here as participatory planning 'tools', were selected for their ability to 'give concrete form to abstract ideas and open processes up to a range of people' (Walsh and Mitchell, 2002, p. 34). Such tools have the potential to support mutual learning, enhance capacity and decision-making (Castleden et al., 2008), and enable cross-cultural communication (Baker et al., 2003). In particular, these tools have been used by Indigenous communities in Australia to develop contemporary natural resource management strategies over the past two decades (Walsh and Mitchell, 2002). However we are not aware of any detailed empirical studies on their direct application in water planning.

Effective participatory tools have the potential to validate community knowledge, heighten ownership of shared information, strengthen relationships between participants, and increase participation and confidence in planning processes (Gambold, 2001; Kyem and Saku, 2009). Using tools either together or in sequence can increase the intensity of participation and the validity of planning outcomes by offering overlapping approaches to authenticate information (Broughton and Hampshire, 1997, p. 156).

From early observations and discussions with the TLC and the Water Planner, we identified a need to both raise awareness of water issues and develop ways of providing a forum where both community members and their institutions could engage in the planning process. This forum should facilitate discussion on a range of matters which included water values, current and planned water uses, and objectives for water resource management. The tools also had to be flexible enough in their design and delivery to accommodate diverse interests in water resources: different methods of engagement might be needed to accommodate different groups, for example participants with different levels of knowledge and technical expertise in Western water resource management. In conformity with cultural protocols, we consulted with the TLC and there was no requirement to hold separate forums to engage women. However as the project developed we surmised that many women found it more difficult than men to attend weekday workshops and thus adjusted our methods.

In negotiation with NRETAS, the Water Planner and TLC officers, we developed four main tools. They are:

1. Community workshops featuring reconnaissance trips to water infrastructure and water places of local significance.
2. Participatory community mapping to elicit current water uses and values and share information on existing management practices.
3. An operating 3D Physical Groundwater Model relating groundwater, rainfall, aquifer recharge, production bores, billabong and spring flow.
4. Visits-to-Country with Tiwi Water Trustees, senior land owners who have been nominated to represent the interests of each land owning group, to identify significant water places, elicit water values and aspirations, and discuss approaches to the development of a Water Management Strategy.

4.1. Community workshops

Community workshops are a flexible and adaptable participatory planning tool as they allow for different learning styles, and tailor activities to meet varied interests (Whiting et al., 2007; Live & Learn Environmental Education, 2007; Walsh and Mitchell, 2002). The Water Planner and WPT researchers attended two community gatherings and two meetings of the Tiwi Land Council before running a trial community water planning workshop with the Tiwi Rangers. Following the trial workshop, four community workshops were carried out with several main objectives: to raise awareness and share information about Tiwi water resources; and to discuss the concept of Western scientific water management and the potential elements of a water plan for the Tiwi Islands. They were held between November 2009 and April 2010 in the main communities of Pirlangimpi, Milikapiti, Ranku and Nguiu. The number of people attending the workshops varied from 10 to 40 and included children.

We designed an evaluation process for the workshops focused on both content and process. The first two workshop evaluations found that groundwater issues seemed to have little relevance to the participants who were more concerned about water supply and infrastructure. To introduce discussion on groundwater resources we used a working 3D Groundwater Model to link water resources to water supply.

As the WPT project progressed, we adapted the workshop format and content to encourage participation, and improve relevance. For example, we learnt that the workshop venue was important. Workshops held indoors attracted fewer participants and generated little discussion in comparison with later workshops held in open-air buildings and outside. At the first workshop, we observed that the local site visit or 'reconnaissance trip' (Community Planning Net, 2012) stimulated the most active participation. Reconnaissance trips, a recognized planning tool, involve a group of local knowledge experts and technical experts traveling together to places to identify key issues and review progress in planning. By 'planning tool' we are referring to an approach or method to support community engagement in the water planning process. The aim of these trips was to familiarize everyone with the physical environment and promote a discussion of key issues. During the trips participants interacted informally as they visited sites including: natural water features; water supply and monitoring infrastructure (i.e. bores); and water users (for example a community horticulture enterprise). Participants volunteered questions or responded to questions and prompts from the workshop facilitators.

Subsequent workshops took place outdoors during the reconnaissance trips. This allowed the process to be as experiential as possible and to allow collective activities to give rise to questions and input. A sense of joint inquiry was enhanced by the Water Planner's demonstration and explanation of conceptual elements at particular water sites. For example, participants visited a bore and discussed groundwater level fluctuations and recharge. They also visited a local spring-fed swimming hole and discussed water-dependent ecosystems and the Western scientific concept of the water cycle. Visits to the water pumping station produced discussions on chlorination, sewage treatment issues and pumping and control mechanisms. These on-site discussions served to share and align understandings between people with different perspectives and knowledge of Tiwi water resources and management. In other research such discussions have shown improved communication and local management effectiveness (Abel et al., 1998).

4.2. Participatory mapping

Participatory mapping (Muller and Wode, 2003; IFAD, 2009; Gambold, 2001) was used in two of the community workshops. This tool features visual rather than verbal analytical and presentational techniques to elicit, collect and plot community information on the location, access and use of resources, in this case, water resources within a community. The United Nations Environment Programme describes Participatory (Resource) Mapping as 'an emerging tool to empower local communities and Indigenous peoples to become more involved in natural resource management and environmental protection' (Mbile, 2008). The tool has been previously used in Australia to represent institutional constructs such as relationships between different organizations or groups, and record topographical phenomena from remote and sparsely populated regions (Walsh and Mitchell, 2002). Walsh maintains it resembles 'traditional Aboriginal methods for describing Dreaming tracks and sites' and is therefore easily understood by local people (Walsh and Mitchell, 2002, p. 135).

For the WPT workshops we used large-scale maps and aerial photographs to stimulate community discussions on water resources location, uses and values. At the billabong in Pirlangimpi the researcher and Water Planner used copies of a map of the area to focus discussion on past uses and contemporary management concerns. In considering the map together with the researchers the women provided the following information:

- the billabong is an important cultural site for Tiwi Islanders;
- they used to fish and camp at the billabong as children;
- a large crocodile is believed to inhabit the billabong.
- concerns about potential salt water intrusion on the seaward side of the billabong in the event of sea level rise.

At the Ranku community meeting of both men and women we employed maps and a laminated A3-sized aerial photograph of the region as a template on which to locate and record information. While the male Water Planner located water sites on the map with the men's group of nine participants and discussed site attributes and issues, the female WPT researchers worked with a smaller number of women to identify:

- particular wetland areas as good hunting places;
- where the community had drawn its water several months earlier when their bore ceased to operate;
- the type of bush tucker or wild food resources available in each place according to the time of the year, water and environmental conditions and
- convenient waterholes for collecting clean water, favorite swimming holes, and areas for camping, and water-related concerns about those areas.

4.3. Physical 3D Groundwater Model

A working 3D Physical Groundwater Model, as shown in Fig. 2, introduced after the first workshop, was a major, well-received component of the remaining community workshops. It demonstrates a cross-section of an operating production bore and its aquifer and simulates rainfall, aquifer recharge, pumping from a production bore, and billabong and spring flow. At community workshops the Water Planner demonstrated the model to simulate hydrologic features and processes and followed this with a discussion of the hydrologic cycle (ground and surface water interactions) and geo-morphological processes, and used a schematic of the water cycle from a workshop pictorial booklet. Typically, the demonstration of the aquifer recharge and discharge cycle was repeated several times. Workshop participants were interested and curious about the physical model, and they developed an understanding of the model especially with respect to each rainfall and pumping event, the spring flowing as a result of a high and rising water table, and the drying out of the billabong as the water table lowered.

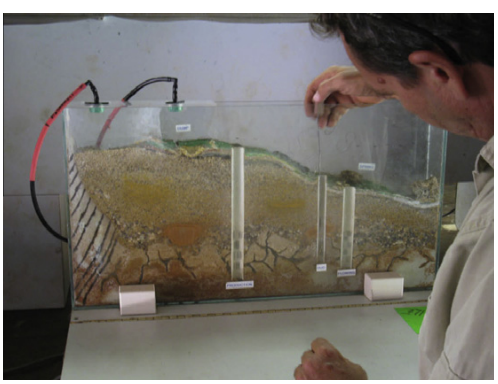

Fig. 2. The Physical 3D Groundwater Model (photo: Ayre, 2009).

The objective of using the Groundwater Model was to use a tangible and visual representation of water dynamics as a catalyst to promote an exchange of knowledge and to stimulate discussion. This was successful possibly because it was neither reliant on written information nor on complex technologies. It created a shared experience for the participants in which they could initiate discussion about changes in their own local water features over the wet and dry seasons. In some workshops they considered the potential effects of changing rainfall patterns from climate change or the impacts of increasing water demand.

4.4. Visits-to-Country

To ensure that the planning process met the needs and aspirations of the TLC and other Tiwi organizations we introduced Visits-to-Country based on the 'reconnaissance trip'. The Trustees were asked by WPT project researchers to identify the location of 'important' water sites on Country. As discussed below in Section 5.1 the Trustees initially took responsibility to take the Water Planner and researchers to these locations. As governance issues were resolved the Trustees delegated the visits to the Tiwi Rangers.

This research demonstrates the importance of the Visits-to-Country planning tool in enabling the appropriate participation of Tiwi knowledge authorities and traditional owners in water planning. There were three main aspects to the success of this tool. Firstly, Trustees hold particular native title and custodial rights and responsibilities on behalf of their kin to care for and manage particular tracts of lands and waters or 'Country'. This responsibility is often referred to by Australian Indigenous peoples as 'speaking for Country'. On the Tiwi Islands, the Trustees' responsibilities include the right to make decisions about water resources and to represent others who have ownership and custodial interests in particular custodial estates or Country. It is important to recognize this key role of Trustees and to formally acknowledge their unique contribution of significant and specialized knowledge (IIRR, 1996).

Secondly, the TLC supported the 'Visits-to-Country' because Trustees need to be knowledgeable about and engaged in the water planning process to be able to provide advice and information to the community.

Thirdly, the learning environment created through the 'Visits-to-Country' planning tool supported the integration of Western and Tiwi knowledges through a shared experience of being together on Country. Bosch et al. (2003) noted that managers and scientists view landscapes and interpret landscape processes differently from community (see also Ross and Abel, 2000). Non-Tiwi and Tiwi engaging together through the Visits-to-Country found ways through joint inquiry and reflection to mutually translate understandings of landscape processes and water knowledge *in place*. For example, the Trustees and Rangers could instruct the WPT researchers and the Water Planner about the value of particular water sites and their aspirations for water management; and the landowners could ask about matters such as the scientific assessments of Tiwi water resources and the planning process. Visits-to-Country therefore promoted relationship building and mutual learning. This mutual engagement helped the parties identify and record water sites of value; assess, confirm or select new sites for water monitoring; examine the condition of water monitoring infrastructure, and undertake water sampling. As a direct outcome of the Visits-to-Country, we recorded Tiwi knowledge and perspectives on water resources and, along with the Water Planner, learnt about Tiwi water values and management objectives.

5. Results and outcomes

5.1. Adaptive approach

The arrangements for participation needed to satisfy both formal government water planning protocols and Indigenous governance and decision-making protocols. Our research demonstrates that a flexible approach to engagement is required. To give an example, in March 2009 it was necessary to make a decision regarding membership and roles in a proposed consultative body for the water planning process. The TLC established a Tiwi Water Advisory Group (TWAG) and appointed the Trustees to this group. Two months later however the TLC decided that existing mechanisms provided adequate opportunities for consultation on water issues and that senior Tiwi decision-makers should be engaged directly through Visits-to-Country. However only two Visits-to-Country were made before the Trustees delegated this task to the Rangers. Subsequently the Tiwi NRM Committee took over the role of the TWAG. Two months later the NRM Committee recommended that a small working sub-committee made up of Rangers should work with the Water Planner and the WPT researcher as part of the water resource strategy. The Rangers would develop options and bring proposals to TWAG for agreement and subsequent TLC endorsement.

These changes reflected Indigenous decision-making with regard to accountability. According to Kate Hadden, Environment Secretary for the TLC, nominating the 'Trustees as the first "TWAG" acknowledges their position and authority for Country. This is critical for appropriate consultation and shows the broader Tiwi community that processes are occurring in the correct hierarchy' (Kate Hadden, pers. comm., 2009). Successive decisions devolving responsibility for the operational role through the hierarchies down to the subcommittee of Rangers: '…are evidence that each "level" became confident enough in the project to devolve these responsibilities down to the "nuts and bolts" level' (Kate Hadden, pers. comm., 2009).

5.2. Evaluation of tools

The planning workshops were evaluated by both the project team and the participants through discussions at the end of workshops. Subsequent modifications improved the content, methods and relevance of the participatory planning tools. A formal survey of the community engagement processes (using a differential format of tick boxes, with a Likert scale from 1–5 with 1 indicating 'not useful' to 5 indicate 'useful all or almost all the time') was also completed by the Rangers, the only group to have participated in the repeated use of all tools. Rangers were asked to assess, first on their own behalf and second on behalf of community participants, how useful each tool had been in engaging them and allowing them to share information about their water issues, uses and values. Responses on behalf of both Rangers and community participants revealed a high level of interest and appreciation for the information provided by the 3D Groundwater Model. The community workshops and supporting posters and workshop booklets tailored to each community were deemed the next most useful engagement tool followed by participatory mapping and Visits-to-Country (possibly because both had only reached a limited audience). Fig. 3 below represents results of a formal evaluation by the Tiwi Rangers of the four planning tools trialed.

The four main planning tools differ in the extent to which they enable the mutual recognition and sharing of Western scientific knowledge and Tiwi knowledge and values. The ideal tool would transmit both Western and Tiwi knowledge and values equally well. Individually our tools did not; though used together the set did deliver a balanced exchange of information. Fig. 4 below depicts this variation.

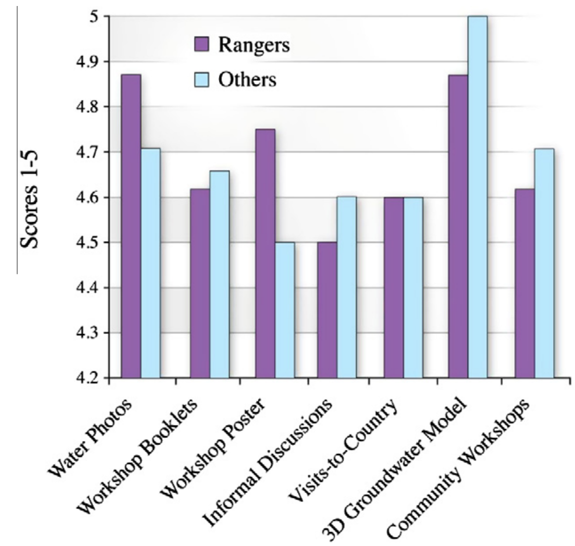

Fig. 3. Tiwi Land Rangers' survey questionnaire evaluation results (Tan et al., 2012, p. 357).

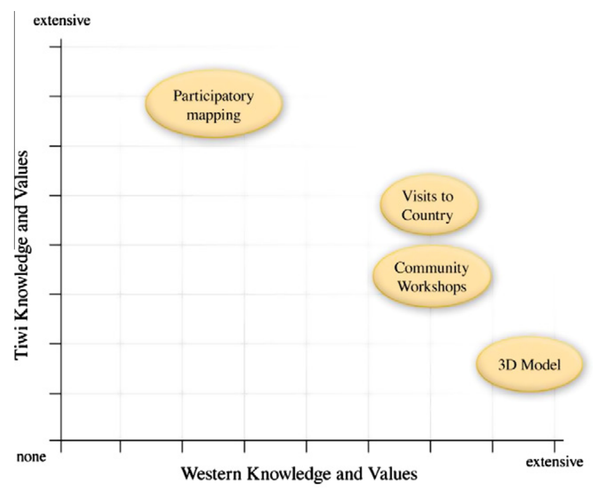

Fig. 4. Transmission effectiveness of participatory planning tools (Tan et al., 2010, p 353).

The relative positions depicted in Fig. 4 above are based on how the tools were actually used, not what their full potential use might deliver. For example, participatory mapping was introduced specifically to elicit and explore Tiwi values in water however it can be used as a tool for other purposes, such as to communicate more information about Western water planning. Likewise, Visits-to-Country could have been used merely as a means of eliciting Tiwi water uses, values and objectives. However we found that working relationships developed and mutual exchanges of values and knowledge with Tiwi water resource managers took place while sharing a vehicle over the course of a day visiting important water sites together.

6. Results and discussion

6.1. Attendance of women

We initially did not provide for separate activities for men and women. However we found that generally women were under-represented in the engagement activities, despite our efforts to ensure women knew of them. For example, during a preliminary trip to Nguiu to confirm venue arrangements and distribute posters, project team members sought out workplaces with high concentrations of women to discuss the up-coming workshop, yet only four women attended. In the more remote communities where there were perhaps fewer opportunities for women to work outside the home, attendance at workshops was more balanced. This may indicate that women who work outside the home find it more difficult than men to attend weekday workshops, and that a weekend hunting and gathering activity, e.g. freshwater crabbing, might possibly create a more convenient opportunity for women to discuss water use and values.

Planning activities that facilitate women's involvement in Indigenous water planning is an important consideration for future engagement as men's and women's knowledge and use of resources may differ significantly (Simpson, 1994; Rambaldi et al., 2006; IK Notes, 2003). Both should be represented in decisions about water resource management.

6.2. High resource requirements

Limited and costly communication and transport facilities, complicated by constrained access arrangements, were key impediments to open communication and good community engagement. These are general concerns in remote areas. Part way through the planning process NRETAS dedicated a vehicle for the Water Planner's use. This considerably improved his and the WPT researchers' travel arrangements.

Another issue was the poor telecommunication coverage in isolated rural communities where few residents own a mobile phone. Asking others to assist in communicating meeting details can be easily thwarted by unanticipated circumstances, and interacting with community members only when an occasion arises can be very resource intensive. A partial solution was to integrate support across the various institutions to share intelligence and tasks, and therefore optimize the use of our networks.

A potential solution to issues of communication and resourcing would be the employment of a Tiwi-based facilitator to assist in logistics and communication. That facilitator would need to be well respected, culturally and socially competent, and adequately resourced. Such a person may very likely already be employed, as we found in the case of our research project.

Good coordination and continuity are critical in achieving effective engagement and can be quite resource intensive. Cost effectiveness of engagement strategies needs to be weighed against the value of successful Indigenous engagement in water planning.

6.3. Matching approaches to circumstances

In many Indigenous cultures complex systems of customary law determine who has access to information and knowledge, and how it can be used. Indigenous knowledge is often privileged knowledge, not freely available but guarded, and not open to inspection or comment by those who do not belong to a particular group or groups (Natcher and Hickey, 2002; Nazarea et al., 1999). This creates practical and philosophical constraints on freely sharing information about water resources. Each water planning process is unique to its context. Our project functioned under a negotiated agreement with the TLC and we sought to undertake this research and the planning process with due respect to and recognition of the unique history, society and circumstances of the Tiwi. We found Tiwi Islanders receptive to new ideas, entrepreneurial, interested in exploring opportunities for economic development and committed to assessing new ideas and ways against their own goals and objectives. We recognize therefore that methods and approaches chosen for engagement, the timeframes required to establish familiarity, and interpersonal and institutional trust may differ in other contexts.

However we can draw several key learnings about the application and characteristics of participatory planning tools which may apply in other contexts where Indigenous interests and values in water are critical in determining water management and allocation regimes.

1. The planning process needs to employ culturally appropriate participatory planning tools to elicit and record knowledge, values and perspectives on water resources and features, to identify sources, estimate levels of future water demands, and explore the cultural, social, economic and environmental impacts of water use and management options.
2. The selection and development of planning tools appropriate to Indigenous contexts must be negotiated between water planners, researchers and Indigenous community members and properly resourced.
3. Formal working arrangements (partnerships) must be established using protocols for the design, implementation and review of participatory planning tools alongside a process for monitoring progress in developing these tools.
4. Positive and productive working relationships between water planning and research agencies and Indigenous communities and their institutions are required to apply appropriate planning tools. Such relationships will be supported by adequate timeframes for mutual learning, adherence to Indigenous community protocols and processes, and sufficient resourcing to support participation in research.
5. Planning tools which support mutual learning are characterized by adequate development time, interactions on Country (e.g. participatory mapping), allowing a diversity of participants to share knowledge (e.g. community workshops), and maintaining continuity of interaction.
6. Planning tools that support participation are characterized by the use of existing networks (e.g. institutions such as the Tiwi Rangers); links to existing activities (e.g. Community Open Days); incentives to participate (e.g. provision of appropriate workshop venues, engaging activities, etc.); and activities that addressed community needs and interests as a priority.
7. Planning tools that support awareness-raising are characterized by material demonstrations and communication (e.g. 3-D Physical Groundwater Model; Tiwi Community Water Planning booklets and posters).
8. The presence of an Indigenous Water Planner who had worked previously in the Tiwi Island was an important factor for success.
9. Methods and approaches that facilitate the participation of Indigenous people in water planning must articulate the clear benefits of water planning to the population. Without this, participation will be seen as a waste of time.

6.4. Outcomes to date

The Draft Tiwi Water Resource Strategy was submitted to the Department (NRETAS) and the Minister for his/her approval in 2011. Some delays in releasing the Draft Strategy for comprehensive consultation with Tiwi Islanders stakeholders occurred due to accountability procedures (i.e. related to finance, corporate affairs, legal and public affairs) within the NT government. Accommodating both Tiwi and NT government institutional arrangements and governance to achieve mutual goals and outcomes was recognized by officers of NRETAS and the WPT researchers as an important and ongoing challenge of the Tiwi water planning process. An internal, departmental working group (within NRETAS) (C. Wicks, pers. comm., January 2012) to support intra and extra-institutional coordination relating to development of the Tiwi Water Resource Strategy was mooted; however, such a group has not yet been established. A principal objective of this group would be to champion Tiwi water planning within the Department and engender acceptance from internal stakeholders for any 'non-standard approaches' required to credit and acknowledge Tiwi authority and custodianship of Tiwi water resources within the NT water planning framework and other governance regimes.

The Secretariat of the Tiwi Land Council noted that the process of developing the draft Strategy has been successful to date with the efforts and commitment of the Water Planner and the WPT project contributing very positively to the process (K. Hadden, pers. comm., January 2012). The Water Planner has commented that Tiwi people: '... recognize the strategy as a document of which they have ownership and identifies their views on how they would like to manage water resources on the [Tiwi] Islands.' (M. Schmid, pers. comm., January 2012). He noted that Tiwi people involved closely in the process have informed him that they have '...learnt a lot from the process and it has influenced them in thinking more about their futures particularly from a natural resource management perspective' (M. Schmid, pers. comm., January 2012). The Water Planner also noted that providing a number of options for engaging members of Tiwi communities through the WPT project was an important part of achieving ownership of and support for the process on the part of Tiwi people and their communities.

7. Conclusion

In normative water planning processes in Australia water advisory committees are asked to consider key issues and pre-considered options for water management generally based on an understanding of benefits to particular segments of the population. This understanding is principally derived from a Western conception of water and its values; it does not usually take account of the range of Indigenous values in water. With the aim of acknowledging and incorporating the varied interests and values held by Tiwi Islanders this research developed, in collaboration with both the government planning agency and Tiwi institutions and communities, mechanisms and approaches (participatory 'planning tools') to promote and facilitate engagement of Tiwi Islanders and their institutions in decisions about the management of their water resources. This research therefore provides important insights and lessons on how to engage Indigenous water managers in planning processes and contributes to a growing body of Australian (Baker et al., 2003; Altman and Jackson, 2008; Moorcroft et al., 2012; Ens et al., 2012;) and international literature (Borrini-Feyerabend et al., 2004; Berkes, 2008) on cross-cultural approaches to managing Indigenous 'country' or custodial estates.

Recent feedback from Tiwi suggests that participatory methods and approaches can improve Indigenous engagement in water planning: when the Draft Tiwi Water Resource Management Strategy was circulated for comment, the TLC and the general community found no contentious issues and attributed this to the approach taken in the WPT project (Kate Hadden, pers. comm., December, 2011).

However responding to the challenge of Indigenous engagement in water planning is not merely a matter of refining the standard water planning approach of providing information, listing key issues, and gathering feedback on pre-considered options (Tan et al., 2010). We suggest that planning agencies must consider the unique context and needs of Indigenous communities. The use of appropriately tailored and resourced planning tools will support both agencies and Indigenous communities in undertaking water planning.

Establishing institutional commitment to Indigenous engagement is important, a factor that became apparent after the conclusion of the research. Smooth acceptance by internal (government) departmental stakeholders is at issue when procedures challenge or even just fall outside standard operating processes. Thus any 'creative approaches' that acknowledge Indigenous authority and ownership will need to overcome potential hurdles within the institution's finance, corporate and legal arms. An internal working group within government is one way to address this.

Good engagement using appropriate planning methods and approaches is needed to ensure the genuine participation of Indigenous people and their institutions in the water management reform in Australia prescribed by the NWI. Understanding what Indigenous communities and institutions deem beneficial and valuable about water use and management will require increased effort on the part of government planning agencies as it cannot be assumed to be identical to other stakeholders' needs.

Unless meaningful dialogue takes place with Indigenous stakeholders to identify the water values, uses and aspirations of their communities and institutions, much of the formal water reform process in Australia will continue to be perceived as irrelevant to Indigenous peoples.

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References

Abel, N., Ross, H., Walker, P., 1998. Mental models in rangeland research, communication and management. Rangeland J. 20 (1), 77–91.

Altman, J., Jackson, S., 2008. Indigenous land and sea management: recognize, respect and resource. In: Lindenmayer, D., Dovers, S., Harriss Olson, M., Morton, S. (Eds.), Ten Commitments: Reshaping the Lucky Country's Environment. CSIRO Publishing, Collingwood, pp. 207–214.

Ayre, M., 2009. Context, Issues and Gap Analysis for Tiwi Islands, Northern Territory. Water Planning Tools Project Report, National Water Commission Canberra. <http://www.waterplanning.org.au> (accessed 13.03.11).

Baker, R., Davies, J., Young, E., 2003. Sharing knowledge (introduction to Part III). In: Baker, R., Davies, J., Young, E. (Eds.), Working on Country: Contemporary Indigenous Management of Australia's Lands and Coastal Regions. Oxford University Press, South Melbourne.

Berkes, F., 2008. Sacred Ecology: Traditional Ecological Knowledge and Management Systems. Taylor and Francis, London.

Bhattarya, S., Nitesh, T., 2004. Integrating indigenous knowledge and GIS for participatory natural resource management: state-of-the-practice. Electr. J. Inform. Syst. Develop. Countries 17 (3), 1–13.

Borrini-Feyeraband, G., Pimbert, M., Farwar, M.T., Kothari, A., Renard, Y., 2004. Sharing Power: Learning by Doing in Co-management of Natural Resources Throughout the World. IIED and IUCN/CMWG, Cenesta, Tehran.

Bosch, O., Ross, H., Beeton, R., 2003. Integrating science and management through collaborative learning and better information management. Syst. Res. Behav. Sci. 20, 107–118.

Broughton, B., Hampshire, J., 1997. Bridging the Gap: A Guide to Monitoring and Evaluating Development Projects. Australian Council for Overseas Aid, Canberra, ACT.

Castleden, H., Garvin, T., Huu-ay-aht First Nation, 2008. Modifying photovoice for community-based participatory Indigenous research. Soc. Sci. Med. 66, 1393–1405.

Charles Darwin University, 2008. Interactive Maps and Spacial Interfaces. <http://www.cdu.edu.au/centres/ik/db_flashmap.html> (accessed 24.04.09).

Commonwealth of Australia and the Governments of New South Wales, Victoria, Queensland, South Australia, the Australian Capital Territory and the Northern Territory (COAG), 2004. Intergovernmental Agreement on a National Water Initiative, Council of Australian Governments, Canberra.

Community Planning Net, 2012. Community Planning Methods. <http://www.communityplanning.net/methods/reconnaissance_trip.php> (accessed 12.02.12).

Corbett, J., Rambaldi, G., Kyem, P., Weiner, D., Olson, R., Muchemi, J., McCall, M., Chambers, R., 2006. Overview: mapping for change: the emergence of a new practic. Participat. Learn. Action 54, 1–13.

Davies, J., White, J., Alyson, W., Maru, Y., LaFlamme, M., 2008. Applying the sustainable livelihoods approach in Australian desert aboriginal development. Rangeland J. 30, 55–60 (!).

Ens, E.J., Finlayson, M., Preuss, K., Jackson, S., Holcombe, S., 2012. Australian approaches for managing 'country' using indigenous and non-indigenous knowledge. Ecol. Manage. Restor. 13 (1), 100–107.

Gambold, N., 2001. Integrating perceptions of country through mapping. In: Baker, R., Davies, J., Young, E. (Eds.), Working on Country: Comtemporary Indigenous Management of Australia's Lands and Coastal Regions. Oxford University Press, South Melbourne.

Gray, P., Paiva, J., 2010. Tiwi Islands Water Resource Assessment. Department of Natural Resources, Environment, The Arts and Sport, Darwin NT.

Haig, T., Knapton, A., Matsuyama, A., 2003. Water Resources of Tiwi Islands, Main report, Tiwi Islands Water Study, Department of Infrastructure, Planning and Environment, Natural Systems Division, Darwin, Northern Territory.

Hamstead, M., Baldwin, C., O'Keefe, V., 2008. Water Allocation Planning in Australia: Current Practices and Lessons Learned. Waterlines Report. National Water Commission, Canberra.

Harmsworth, G., 1998. Indigneous values and GIS: a method and a framework. Indigenous Knowledge Develop. Monitor, 6.

Hiscock, P., 2008. Archaeology of Ancient Australia. Routledge, London.

IIRR (International Institute of Rural Reconstruction), 1996. Recording and Using Indigenous Knowledge: A Manual. IIRR, Silang, Cavite, Philippines.

IK Notes, 2003. Women & Indigenous Knowledge: A South–South Perspective Africa Region's Knowledge and Learning Center. <http://www.worldbank.org/afr/ik/default.htm> (accessed 25.03.11).

International Fund for Agricultural Development, 2009. Good Practices in Participatory Mapping: A Review Prepared for the International Fund for Agricultural Development. <http://www.ifad.org/pub/map/PM_web.pdf> (accessed 19.05.10).

Jackson, S., 2005. Indigenous values and water resource management: a case study from the northern territory. Australasian J. Environ. Manage. 12, 136–146.

Jackson, S. (Ed.), 2006. Recognising and Protecting Indigenous Values in Water Resource Management: A Report from a Workshop Held at CSIRO in Darwin, NT, 5–6 April 2006, CSIRO Sustainable Ecosystems, Darwin.

Jackson, S., 2009. Background Paper on Indigenous Participation in Water Planning and Access to Water. Paper Prepared for the National Water Commission. CSIRO Sustainable Ecosystems, Darwin.

Jackson, S., Altman, J., 2009. Indigenous rights and water policy: perspectives from tropical Northern Australia [online]. Australian Indigenous Law Review 13(1), 27–48. <http://www.search.informit.com.au/documentSummary; dn=2504151 68543105;res=IELHSS>ISSN: 1835-0186> (accessed 02.05.11).

Jackson, S., Morrison, J., 2007. Indigenous perspectives in water management, reforms and implementation. In: Hussey, K., Dovers, S. (Eds.), Managing Water for Australia: The Social and Institutional Challenges. CSIRO Publishing, Collingwood.

Jackson, S., O'Leary, P., 2006. Indigenous Interests in Tropical Rivers: Research & Management Issues a Scoping Study for Land & Water Australia's Tropical Rivers Program. CSIRO Sustainable Ecosystems, Canberra.

Jackson, S., Tan, P., Altman, J., 2009. Indigenous Fresh Water Planning Forum: Proceedings, Outcomes and Recommendations. National Water Commission, Canberra.

Jackson, S., Moggridge, B., Robinson, C., 2010. Summary of the Scoping Study: Effects of Change in Water Availability on Indigenous People of the Murray-Darling Basin. CSIRO Water for a Healthy Country Flagship Report. CSIRO, Brisbane.

Kyem, A., Saku, J., 2009. Web-based GIS and the future of participatory GIS applications within local and Indigenous communities. Electr. J. Inform. Syst. Develop. Countries 38, 1–16.

Lane, M., McDonald, G., 2005. Community-based environmental planning: operational dilemmas, planning principles and possible remedies. J. Environ. Planning Manage. 48, 709–731.

Live & Learn Environmental Education, 2007. The Governing Water Guide. Live & Learn Environmental Education, Suva.

Lynch, A., Fell, D., McIntyre-Tamwoy, S., 2010. Incorporating Indigenous values with 'Western' conservation values in sustainable biodiversity management. Australasian J. Environ. Manage., 17.

Mackenzie, J., Tan, P-L., Hoverman, S., Mooney, C., 2012. The value and limitations of action research methodology. J. Hydrol. (this issue).

Matsuyama, A., Haig, T., 2003. Tiwi Islands Water Study, vol. 2. Water Resources of Tiwi Islands, Traditional Knowledge Report. Northern Territory.

Mbile, P., 2008. Community Mapping in Forest Zones of Cameroon: A Scoping of Cases, Questions and Methods Used in Community Mapping and its Relationship with Tenure Recognition. World Agroforestry Center (ICRAF), Rights and Resources Initiative, April 2008. <http://www.iapad.org/publications/ppgis/pmbile_ community_mapping_cameroon.pdf> (accessed 19.05.10).

McFarlane, B., 2004. The National Water Initiative and Acknowledging Indigenous Interests in Planning. Native Title Tribunal, Canberra.

Moorcroft, H., Ignjic, E., Cowell, S., Goonack, J., Mangolomara, S., Oobagooma, J., Karadada, R., Williams, D., Waina, N., 2012. Conservation planning in a cross-cultural context: the Wunambal Gaambera Healthy Country Project in the Kimberely, Western Australia. Ecol. Manage. Restor. 13 (1), 16–25.

Moran, M.F., 2004. The practice of participatory planning at mapoon aboriginal settlement: towards community control, ownership and autonomy. Inst. Aust. Geograph. 42, 339–355.

Muller, D., Wode, B., 2003. Manual on Participatory Village Mapping Using Photomaps: Trainer Guide. <http://wwwiapad.org/publications/ppgis/participatory_mapping_using_photomaps_ver2.pdf> (accessed 11.05.10).

Natcher, D., Hickey, C., 2002. Putting the community back into community-based resource management: a criteria and indicators approach to sustainability. Human Organization 57, 159–170.

National Water Commission, 2011. The National Water Initiative – Securing Australia's water future. Executive Overview: 2011 Assessment, National Water Commission, Canberra.

Nazarea, V., Rhoades, R., Bontoyan, E., Flora, G., 1999. Defining culturally relevant indicators: what are we waiting for? Human Organization 58, 219–220.

Northern Territory Government, n.d. Tiwi Islands: Sites of Conservation Significance, Department of Natural Resources, Environment, The Arts and Sport, 278–283. <http://www.nt.gov.au/nreta/environment/conservation/pdf/09_tiwi.pdf> (accessed 11.07.11).

PowerWater, 2008. Annual Drinking Water Quality Report 2008: Indigenous Communities of the Northern Territory, Power and Water Corporation, Darwin, Northern Territory.

Puri, S., 2007. Scientific with indigenous knowledge: constructing knowledge alliances for land management in India. Manage. Inform. Syst. Q. 31 (2), 355–379.

Rambaldi, G., Kyem, P., McCall, M., Weiner, D., 2006. Participatory Spatial Information Management and Communication in Developing Countries. The Electronic Journal on Information Systems in Developing Countries. <http://www.ejisdc.org/ojs2/index.php/ejisdc/issue/view/85> (accessed 19.05.10).

Ross, H., Abel, N., 2000. Eliciting mental models of landscape processes: the transect method. In: Moore, G.T., Hund, J., Trevillion, L. (Eds.), Environment-Behaviour Research on the Pacific Rim, Faculty of Architecture. University of Sydney, Sydney, NSW.

Schult, J., 2010. A Review of Northern Territory Government Water Quality Data for the Tiwi Islands, Aquatic Health Unit, Department of Natural Resources, Environment, The Arts and Sport, Darwin, NT.

Simpson, B., 1994. Gender and the social differentiation of knowledge. Indigenous Knowledge Develop. Monitor 2 (3).

Srinivasan, L., 1993. Tools for Community Participation: A Manual for Traing Trainers in Participatory Techniques, PROWWESS/UNDP Technical Series. <http://www.chs.ubc.ca/archives/files/CHS0024.PDF> (accessed 12.01.12).

Tan, P.L., Bowmer, K.H., Baldwin, C., 2012. Continued challenges in the policy and legal framework for collaborative water planning. J. Hydrol. http://dx.doi.org/10.1016/j.jhydrol.2012.02.021.

Tan, P.-L., Mooney, C., White, I., Hoverman, S., Mackenzie, J., Burry, K., Baldwin, C., Bowmer, K., Jackson, S., Ayre, M., George, D., 2010. Tools for Water Planning: Lessons, Gaps and Adoption, Waterlines Report, National Water Commission, Canberra.

Tiwi Land Council, 2009. Environmental Studies and Research Access Agreement Between the Tiwi Land Council and Water for a Healthy Country Flagship (CSIRO). Tiwi Land Council, Darwin, Northern Territory.

Tiwi Land Council, n.d.(a). Natural Resource Management Strategy. <http://www.tiwilandcouncil.com/publications/land.htm> (accessed 30.06.10).

Tiwi Land Council, n.d.(b). Environmental Policy, Tiwi Land Council. <http://esvc001013.wic004ty.server-web.com/documents/resources/Tiwi_Env_policy.pdf> (accessed 30.06.10).

URS, 2007. Public Environmental Report: Aircraft Landing Area at Andranangoo Creek West. URS Australia Pty Ltd., Prepared for Matilda Minerals Ltd., Tiwi Islands, Northern Territory.

Walsh, F., Mitchell, P. (Eds.), 2002. Planning for Country: Cross-Cultural Approaches to Decision-Making on Aboriginal Lands. Jukurrpa Books, Alice Springs.

Wang, C., Burris, M., 1997. Photovoice: concept, methodology and use for participatory needs assessment. Health, Educ. Behav. 24, 369–387.

Whiting, S., Hadden, K., Long, J., Lauder, A., Kleidon, A., Cook, K., 2007. Sea turtle Conservation and Education on the Tiwi Islands, Final NHT Report, Australian Government Department of the Environment, Water, Heritage and the Arts, Canberra.

Wollenberg, E., Edmunds, D., Buck, L., 2000. Anticipating Change: Scenarios as a Tool for Adaptive Forest Management: A Guide. Center for International Forestry Research, Borgor.